SiO2/MoS2 core-shell structure properties

"what are the relations between the porosity in the SiO2/MoS2 core shell strcuture and the dielectric constant of them and their effect on the TENG application ?" *Generated by LeapSpace*

In SiO_2/MoS_2 core-shell structures, increasing porosity is expected to *lower the dielectric constant* while improving TENG output only up to an optimum; the main benefit comes from added surface area and charge-trapping sites, but too much porosity can weaken the dielectric response and reduce effective charge density. The evidence is indirect for the exact SiO_2/MoS_2 pair, but it is consistent across porous SiO_2, porous ceramics, and porous TENG layers: more porosity generally means lower dielectric constant, while controlled porosity can improve triboelectric performance through higher surface area and interfacial charge trapping [84] [85] [86] [92] [99].

How porosity changes dielectric constant

Porosity trend	Dielectric-constant effect	What it means
Higher porosity	**Lower dielectric constant** because air/voids replace polarizable solid volume [84] [85] [86] [87] [88]	The shell becomes less polarizable overall
More interfaces / thinner insulating shells	**Can raise dielectric constant locally** through Maxwell–Wagner–Sillars polarization at interfaces [89] [90] [91]	More interfacial charge buildup can offset some porosity effects
Excessive porosity / open networks	**Can reduce effective charge storage and stability** even if surface area is high [92] [93] [99]	The structure becomes less able to hold charge uniformly

What this means for a SiO_2/MoS_2 core-shell TENG

- **Porosity helps TENGs when it is controlled**, because porous structures increase effective contact area and create more trap sites for triboelectric charge retention [92] [94] [99].
- **Lower dielectric constant is not automatically bad for TENGs**; in the retrieved TENG evidence, overly low dielectric constant often tracks with oversized pores and poor charge trapping, which reduces output [99].
- **The best TENG behavior comes from a balance**: enough porosity to increase surface area and interfacial polarization, but not so much that the shell becomes too open, too leaky, or mechanically weak [92] [95] [99].

- In a **SiO_2/MoS_2 core-shell**, the porous SiO_2 shell would most plausibly serve as the **dielectric and charge-trapping layer**, while the MoS_2 core/interface adds heterointerface polarization and structured charge confinement; this is consistent with the broader core-shell and porous-dielectric evidence, though not directly measured for this exact pair [89] [90] [91] [96].

Bottom line

For **SiO_2/MoS_2 core-shell TENG materials**, **more porosity usually lowers dielectric constant but can still improve TENG output if it creates accessible trap-rich interfaces**. The practical target is **moderate, well-connected porosity**: enough to boost surface area and interfacial polarization, but not so much that dielectric constant, charge retention, or structural integrity collapse [84] [92] [92] [99].

Confidence level: **Medium** — the porosity–dielectric-constant relationship is strongly supported, and the TENG implications are well supported in porous dielectric systems, but direct measurements on the exact SiO_2/MoS_2 core-shell system were not provided.

References

84. Oliviero, E., Ruault, M.-O., Décamps, B., (...), Collin, S. (2010) Synthesis of mesoporous amorphous silica by Kr and Xe ion implantation: Transmission electron microscopy study of induced nanostructures. *Microporous and Mesoporous Materials*. https://doi.org/10.1016/j.micromeso.2010.02.015

85. Barnana, Hema Dinesh, Tofail, Syed A. M., Roy, Krittish, (...), ul Haq, Ehtsham (2024) Biodielectrics: old wine in a new bottle?. *Frontiers in Bioengineering and Biotechnology*. https://doi.org/10.3389/fbioe.2024.1458668

86. Xia, Y., Zeng, Y.-P., Jiang, D. (2009) Dielectric and mechanical properties of porous Si3N4 ceramics prepared via low temperature sintering. *Ceramics International*. https://doi.org/10.1016/j.ceramint.2008.09.010

87. Zhang, R., Han, B., Fang, D. (2016) New multifunctional porous Yb2SiO5 ceramics prepared by freeze casting. *Ceramics International*. https://doi.org/10.1016/j.ceramint.2015.12.161

88. Fangsen, L., Wancheng, Z., Hanjun, H., (...), Dongmei, Z. (2009) High performance porous Si3N4 ceramics prepared by coated pore-forming agent method. *Ceramics International*. https://doi.org/10.1016/j.ceramint.2009.05.007

89. Vinayasree, S, Nitha, T S, Tiwary, C S, (...), Anantharaman, M R (2018) Magnetically tunable liquid dielectric with giant dielectric permittivity based on core–shell superparamagnetic iron oxide. *Nanotechnology*. https://doi.org/10.1088/1361-6528/aabc4e

90. Taha, M., Al Kiey, S.A., Khalid, A., (...), Abdallah, S. (2026) Engineering the ZnO/SiO2 Interface for tunable optoelectronic and charge transport properties in Core–Shell nanostructures. *Materials Science and Engineering: B*. https://doi.org/10.1016/j.mseb.2026.119671

91. He, D., Wang, Y., Song, S., (...), Deng, Y. (2017) Polymer-based nanocomposites employing Bi2S3@SiO2 nanorods for high dielectric performance: Understanding the role of interfacial polarization in semiconductor-insulator core-shell nanostructure. *Composites Science and Technology*. https://doi.org/10.1016/j.compscitech.2017.08.006

92. Wu, J.M., Chang, C.K., Chang, Y.T. (2016) High-output current density of the triboelectric nanogenerator made from recycling rice husks. *Nano Energy*. https://doi.org/10.1016/j.nanoen.2015.11.014

93. Ji, C., Cai, X., Zhang, Y., (...), Liu, S. (2026) Porosity–property relationships in Siloxane-based low dielectric constant materials: From atomic visualization to performance prediction. *Applied Surface Science*. https://doi.org/10.1016/j.apsusc.2025.164582

94. Duan, Q., Peng, W., He, J., (...), Nie, S. (2023) Rational Design of Advanced Triboelectric Materials for Energy Harvesting and Emerging Applications. *Small Methods*. https://doi.org/10.1002/smtd.202201251

95. Zhou, H., Li, Z., Huang, J., Liu, J. (2012) High-density Co/Al2O3 core-shell nanocrystal memory. *Materials Research Society Symposium Proceedings*. https://doi.org/10.1557/opl.2012.1020

96. Yang, Y., Veser, G. (2021) Exploiting the impact of pore diffusion in core@shell nanocatalysts. *Catalysis Today*. https://doi.org/10.1016/j.cattod.2020.07.067

99. Zhang, J.-H., Li, Y., Du, J., (...), Wang, Q. (2019) Bio-inspired hydrophobic/cancellous/hydrophilic Trimurti PVDF mat-based wearable triboelectric nanogenerator designed by self-assembly of electro-pore-creating. *Nano Energy*. https://doi.org/10.1016/j.nanoen.2019.04.065